Lecture 03

Mass Spectroscopy

PART - I

Basic Principles

- Mass Spectrometry is an analytical technique that is used to
  - Identify unknown compounds
  - Quantify known materials
  - Elucidate the structural and physical properties of ions
- It is a technique associated with very high levels of specificity and sensitivity.
- Analyses can often be accomplished with minute quantities - sometimes requiring less than picogram (10-12 grams) amounts of material.
- Sample is converted into **gaseous ions** where the masses of the ions are measured

- The mass spectrometer is designed to separate gas phase ions according to their m/z (mass to charge ratio) value.

- The "heart" of the mass spectrometer is the analyzer. This element separates the gas phase ions.

- The analyzer uses electrical or magnetic fields, or combination of both, to move the ions from the region where they are produced, to a detector, where they produce a signal which is amplified.

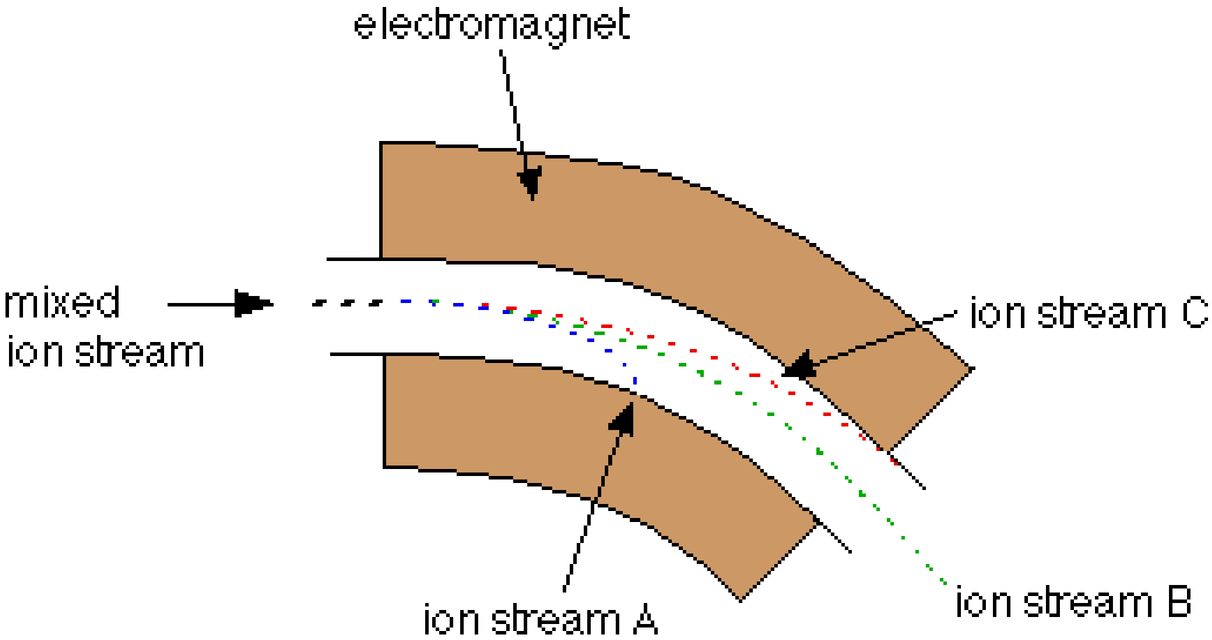
- Since the motion and separation of ions is based on electrical or magnetic fields, it is the mass to charge ratio, and not only the mass, which is of importance.

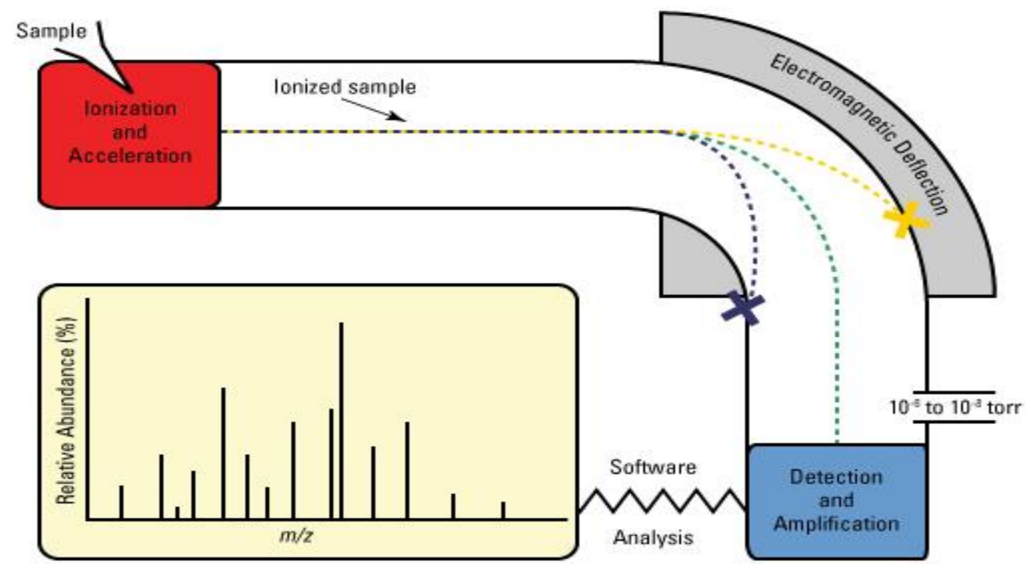
- The analyzer is operated under high vacuum, so that the ions can travel to the detector with a sufficient yield.

Advantages :

1. Very fast analysis < 10 sec.
2. High precision and accuracy
3. Concentration from ppm to %, wide linear dynamic range
4. Cost effective due to high sample throughput

MS Basics


electromagnet
mixed ion stream
ion stream C
ion stream A
ion stream B


Sample
Ionization and Acceleration
Ionized sample
Electromagnetic Deflection
10^{-6} to 10^{-8} torr
Relative Abundance (%)
m/z
Software
Analysis
Detection and Amplification

Ion & Ionization

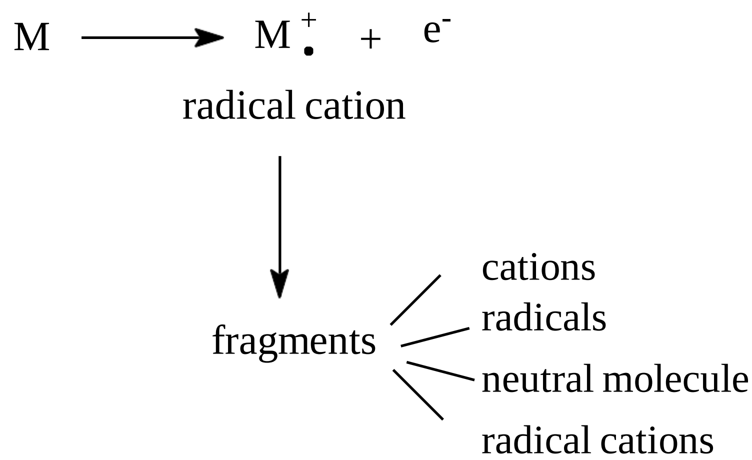
- An ion is a charged particle which results from the addition or removal of electron(s) to a neutral atom or molecule
    - An ion has a mass (m) and a charge (z)
    - Mass is measured in atomic mass unit (a.m.u)

- Ionization is the process where a neutral atom or molecule is converted to an ion

$M + e$	$\Rightarrow$	$M^{\circ -}$	Radical anion
$M - e$	$\Rightarrow$	$M^{\circ +}$	Radical cation
M	$\Rightarrow$	M^{*}	Exited molecule

- vaporized, & bombarded by stream of high energy (70 eV) e^-

- e^- bombardment removes e^- from sample molecule, producing positively charged radical cation called molecular ion, which then fragments further, producing

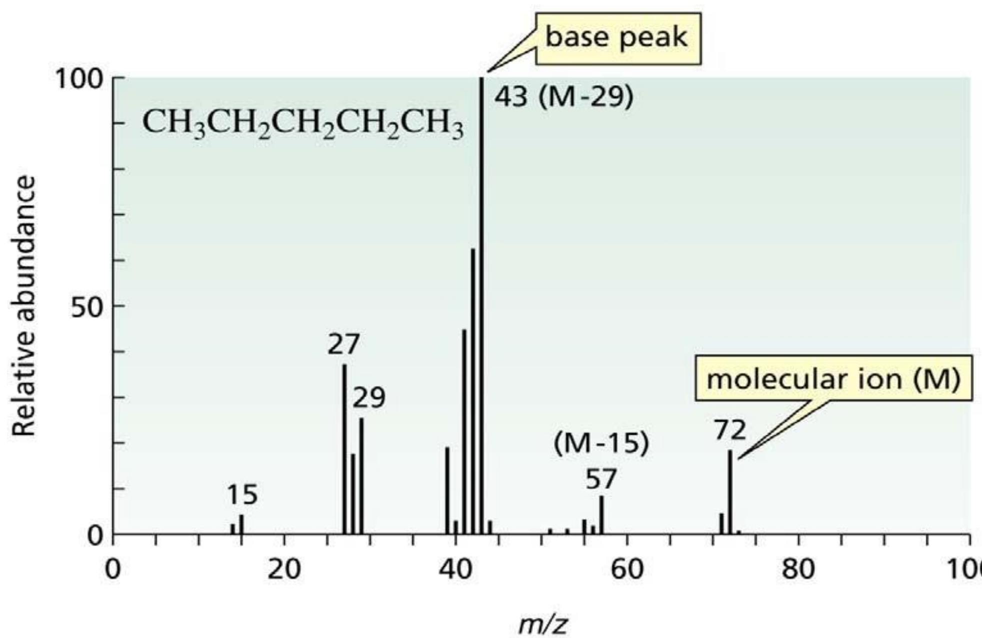


- positively charged fragments, repelled by positively charged repeller plate, pass between two negatively charged focusing & accelerating plates into analyser tube

- magnet surrounding tube produces magnetic field, causing positively charged fragments to move in a curved path; radius of curve determined by mass/charge ratio (***m/z***)

- as charge is constant (e^+), ***m/z*** ratio is molecular mass of fragment

- if fragment path, as determined by *m/z* ratio, matches curvature of analyser tube, fragment passes down tube, through ion exit slit, & strikes collector
- collector records relative number of fragments of that *m/z* passing through slit
- by slowly increasing magnetic field strength, fragments of increasing *m/z* values are guided down tube & through slit onto collector
- plot of relative abundance of each fragment versus its *m/z* value (& ∴ molecular mass) known as mass spectrum (ms)
- only (stable) positively charged particles reach detector & appear on ms



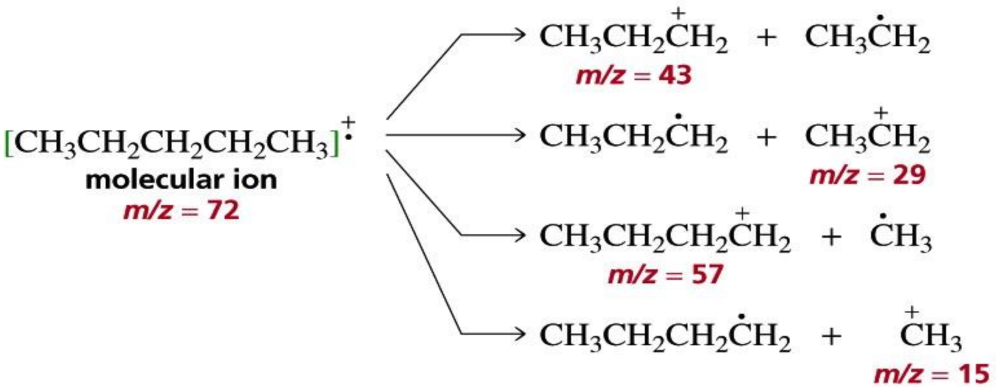
m/z	Relative abundance
73	0.52
72	18.56
71	4.32
57	11.20
43	100.00
42	55.27
41	37.93
39	12.44
29	26.65
28	17.75
27	31.22
15	4.22
14	2.56

- fragmentation pattern determined by
  - bond strength
  - fragment stability

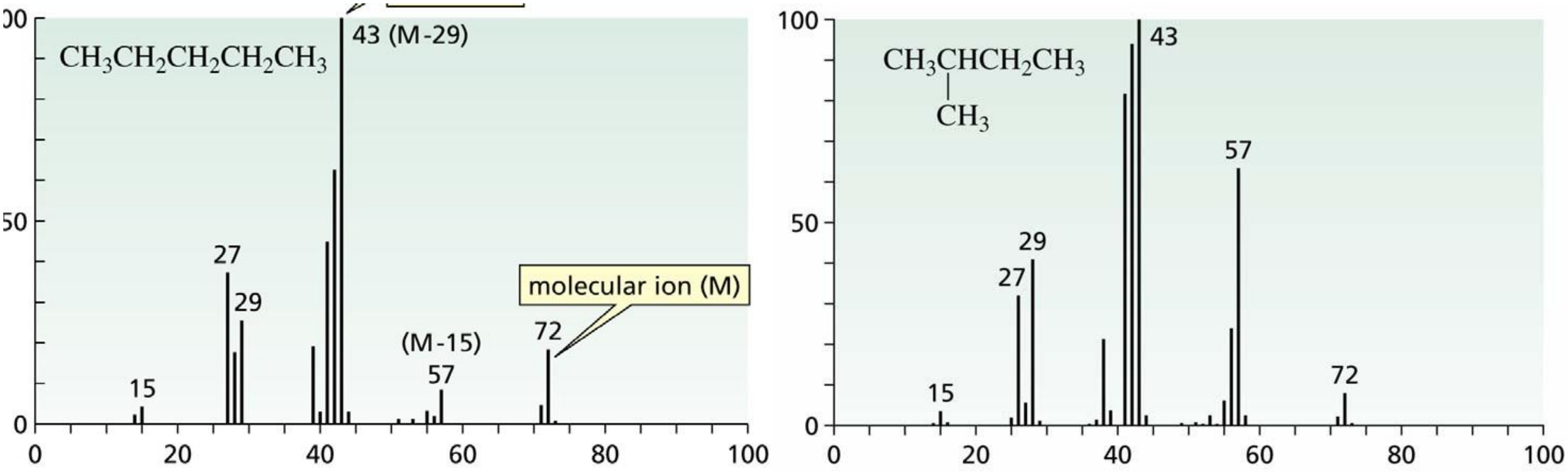
- all C-C bonds in pentane are of ~same strength

- C-2/C-3 bond fragmentation gives 1^{O} C^+ & 1^{O} radical C-1/C-2 bond fragmentation gives 1^{O} C^+ & CH_3 radical
  - 1^{O} radical more stable than methyl radical
  - C-2/C-3 bond more likely to break than C-1/C-2 bond

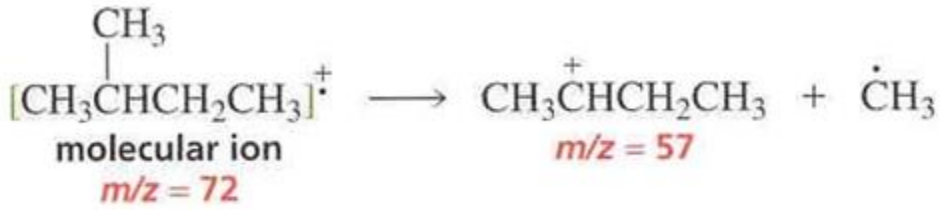
- C-2/C-3 bond fragmentation gives ions of *m/z* 43 & 29 C-1/C-2 bond fragmentation gives ions of *m/z* 57 & 15
  - base peak at *m/z* 43 indicates preference for C-2/C-3 bond fragmentation
  - only (stable) positively charged particles reach detector & appear on ms

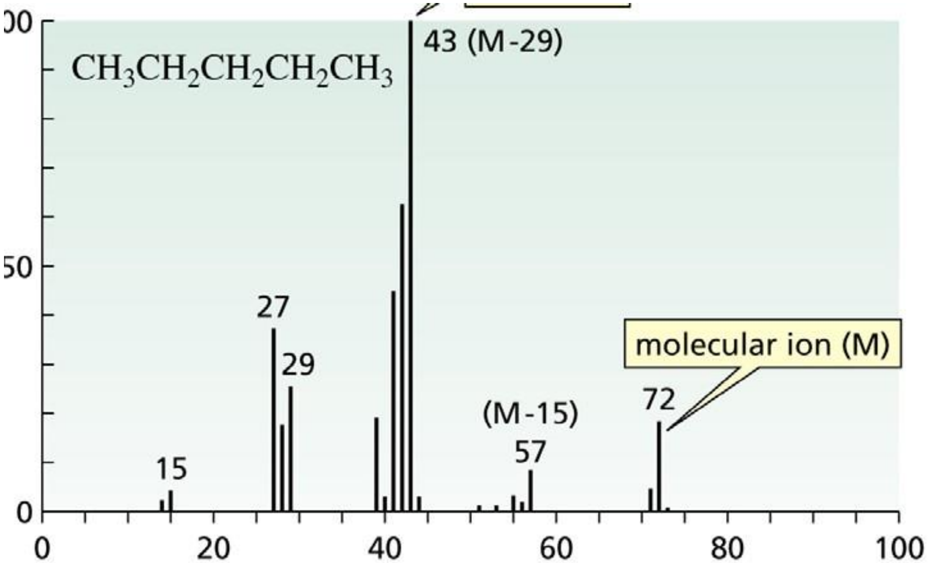


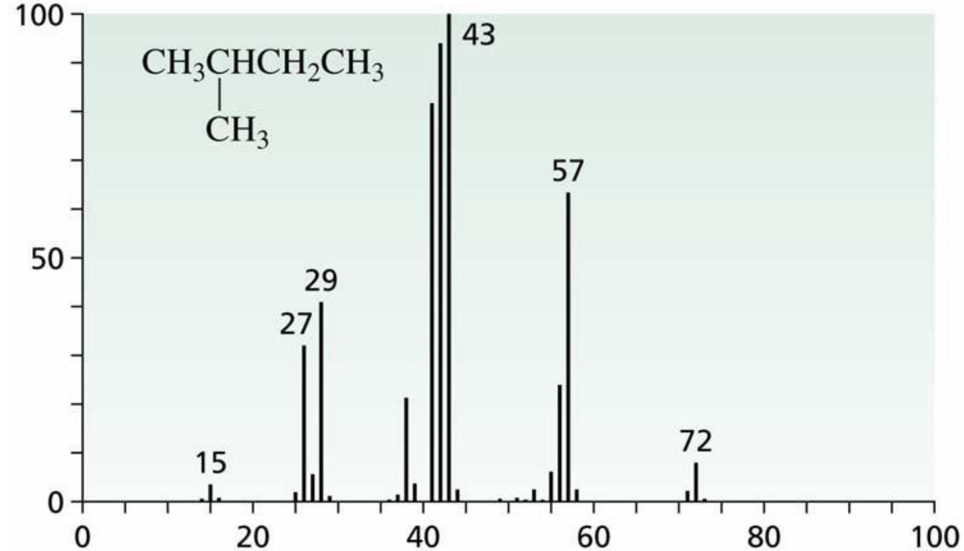
- fragmentation pattern of 2-methylbutane very similar to that of pentane



- single notable exception is intensity of peak at *m/z* 57; is more likely to lose CH_3 radical, as 2^{O} C^{+} formed more stable than 1^{O} C^{+}

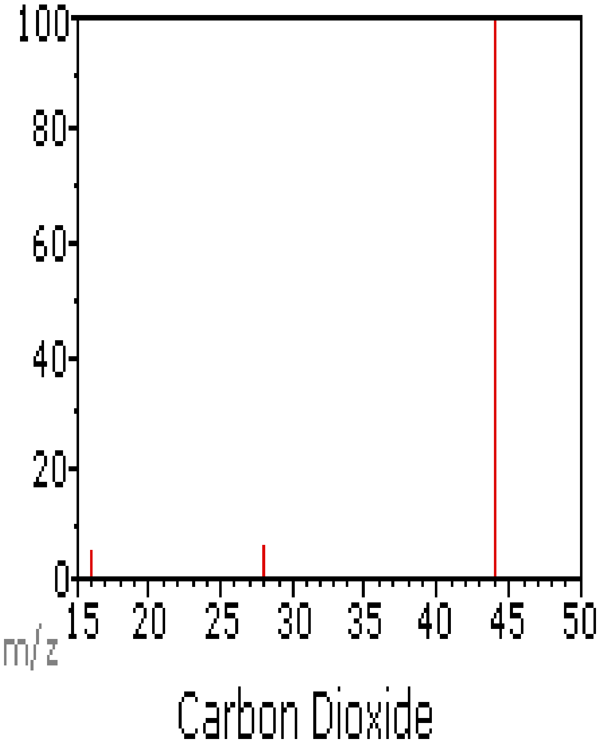


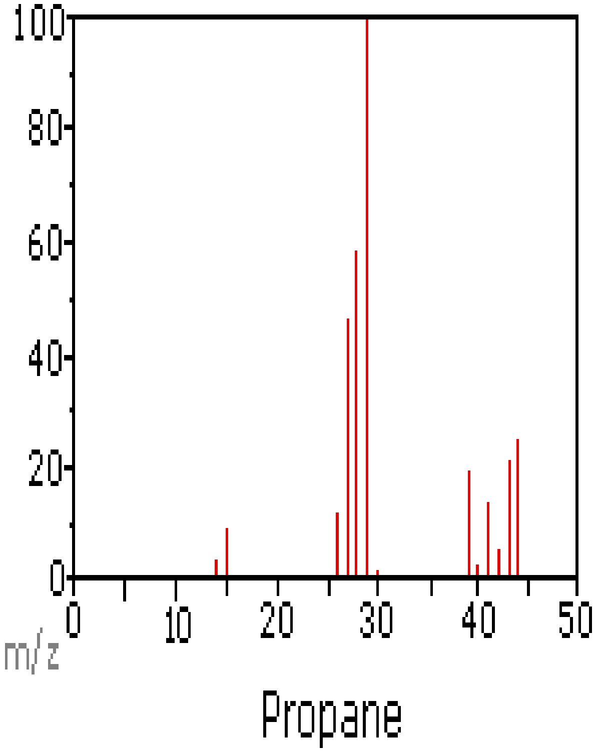


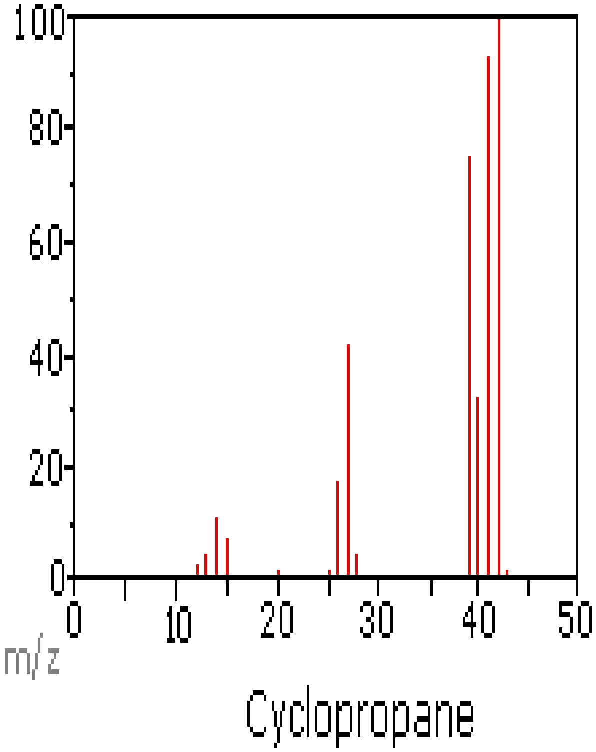


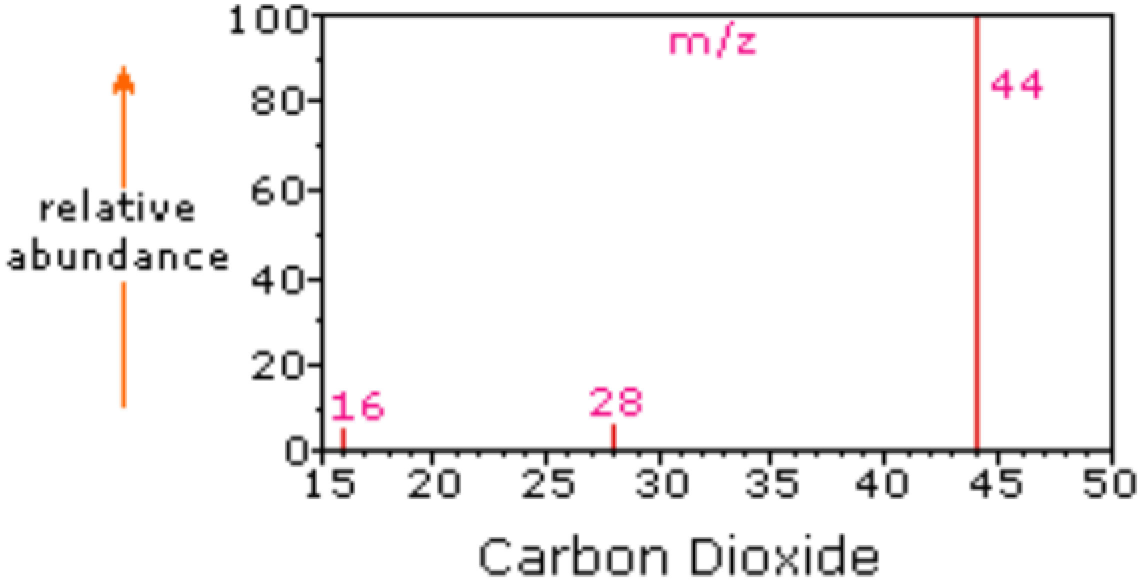
- peaks of 1 or 2 units less than carbocation *m/z* values due to further fragmentation involving loss of H atoms

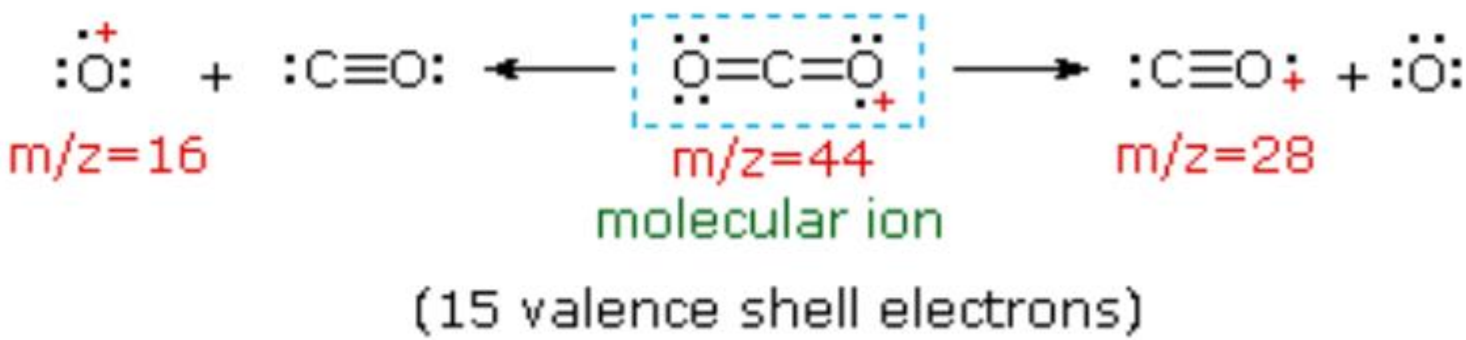
- peak at 1 unit more than molecular ion *m/z* (M+1 peak) due to isotopic effect of ^{13}C
- as 1.11% of naturally occurring C is ^{13}C, relative abundance of M+1 peak in spectrum of compound of formula C_n should be n(1.1)% of molecular ion

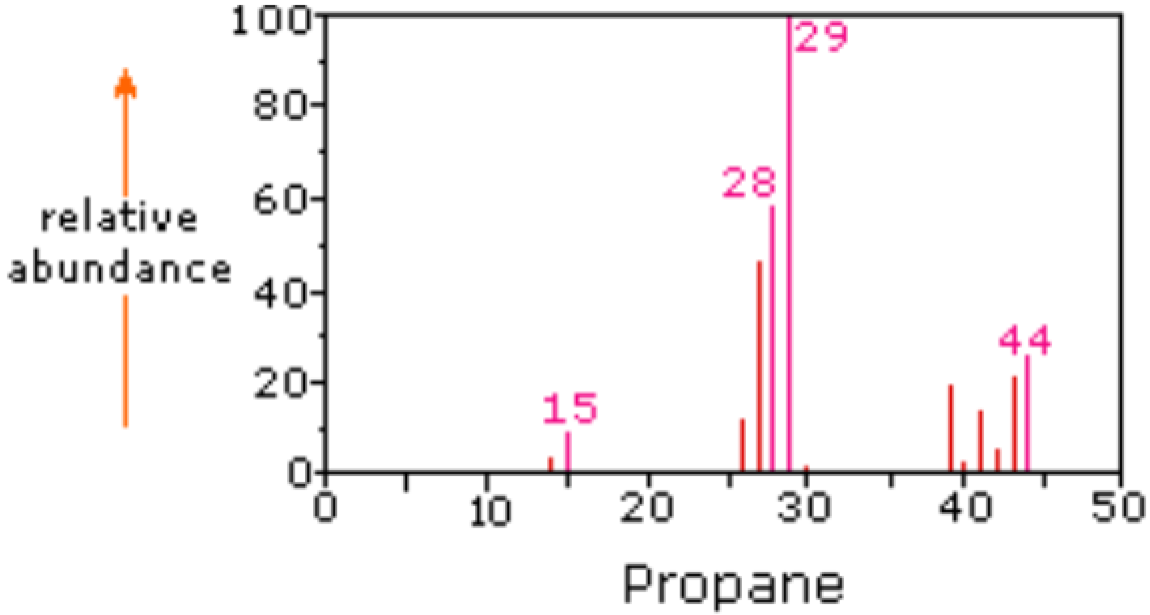

100
80
60
40
20
0
15 20 25 30 35 40 45 50
m/z
Carbon Dioxide

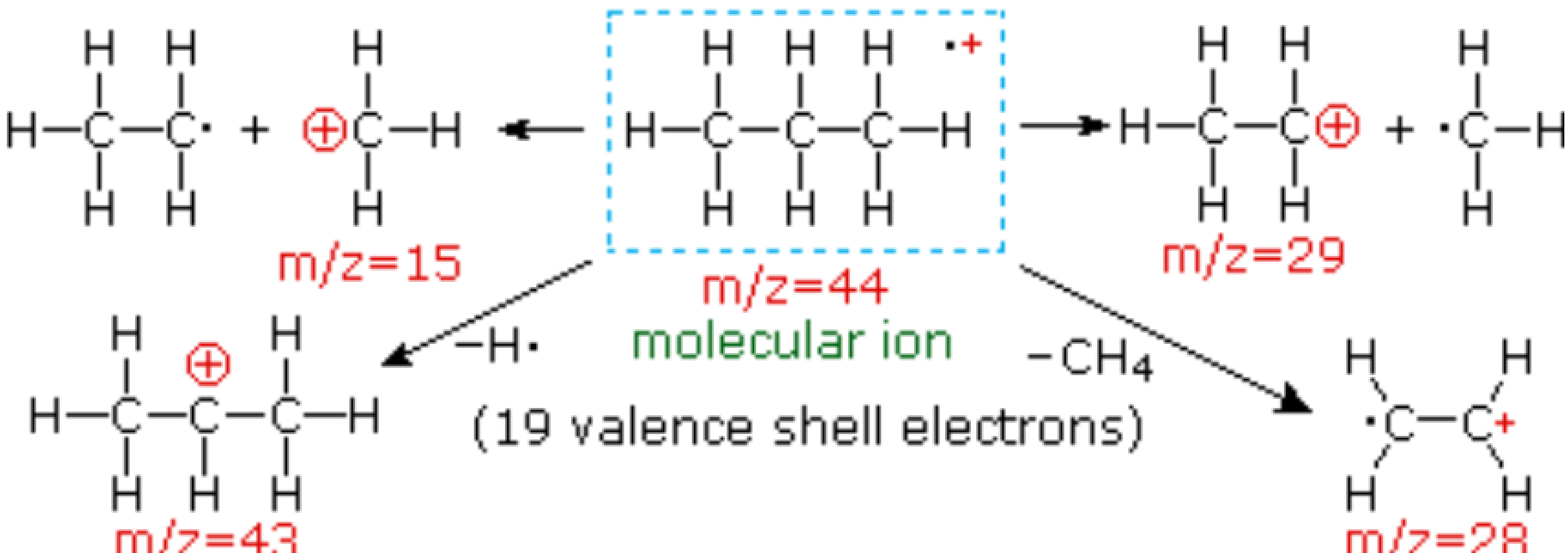

100
80
60
40
20
0
0 10 20 30 40 50
m/z
Propane


100
80
60
40
20
0
0 10 20 30 40 50
m/z
Cyclopropane


relative abundance
m/z
44
16
28
100
80
60
40
20
0
15 20 25 30 35 40 45 50
Carbon Dioxide


m/z=16
m/z=44
molecular ion
m/z=28
(15 valence shell electrons)


100
80
60
40
20
0
relative abundance
29
28
44
15
0
10
20
30
40
50
Propane


m/z=15
m/z=44
molecular ion
(19 valence shell electrons)
m/z=29
−H·
−CH4
m/z=43
m/z=28